

# DATA ANALYSIS REPORT

## Predicting Patient No-Shows in Brazilian Medical Appointments

*An Exploratory Data Analysis of 110,526 Medical Appointment Records*

|               |                                         |
|---------------|-----------------------------------------|
| Dataset       | No-Show Appointments (Kaggle, May 2016) |
| Region        | Vitória, Espírito Santo, Brazil         |
| Records       | 110,526 Appointments                    |
| Analysis Type | Exploratory Data Analysis (EDA)         |
| Date          | 2026                                    |

Edward Amankwah

## 1. Executive Summary

---

Medical appointment no-shows represent a critical operational and financial challenge for healthcare systems globally. In Brazil, where the public health system (Sistema Único de Saúde, or SUS) serves millions of patients, missed appointments lead to wasted clinical resources, increased waiting times for other patients, and deteriorating health outcomes for those who fail to attend. This report presents a comprehensive exploratory data analysis (EDA) of 110,526 medical appointment records collected in Vitória, Espírito Santo, Brazil during 2016, publicly available on Kaggle.

The central objective of this analysis is to identify the patient demographic, socioeconomic, health-related, and scheduling factors most strongly associated with appointment non-attendance. Six research questions were formulated and investigated: the relationship between scheduling lead time and no-show rates; the influence of patient age; the effect of chronic health conditions; the role of SMS reminder messages; the impact of the appointment day of the week; and a synthesised feature-importance ranking across all available predictors.

Key findings establish that scheduling lead time (wait days) is the single strongest predictor of no-shows, with same-day appointments resulting in only a 6% absence rate compared to 45% for appointments scheduled more than 60 days in advance. Patient age, neighbourhood of the clinic, and enrolment in the Bolsa Família welfare program (Scholarship) also demonstrate meaningful associations. Contrary to expectations, patients with chronic conditions such as hypertension and diabetes are more reliable attenders, while the raw effect of SMS reminders is confounded by wait time. Gender and alcoholism exhibit negligible predictive value in this dataset.

These findings carry practical implications for healthcare administrators seeking to allocate resources efficiently, design targeted reminder systems, and reduce the systemic costs of no-shows. All results are presented as associational and exploratory; causal

claims are explicitly avoided. The report concludes with a limitations analysis, directions for future research, and a value statement articulating the business and societal case for acting on these insights.

Dataset: No-Show Appointments — Kaggle (May 2016)

Records Analysed: 110,526 appointments (1 row removed: invalid age)

Overall No-Show Rate: 20.2%

Top Predictor: Wait Days between scheduling and appointment ( $|r| = 0.186$ )

Primary Tool: Python (Pandas, NumPy, Matplotlib, Seaborn)

## 2. Problem Statement

---

Missed medical appointments — commonly referred to as no-shows — are a pervasive and costly phenomenon in healthcare systems worldwide. When a patient fails to attend a scheduled appointment without prior cancellation, the consequences ripple through the entire healthcare delivery chain: appointment slots go unused, clinician time is wasted, patients who need care cannot access those freed slots in time, and overall system efficiency declines. In the Brazilian public health context, where the SUS must balance high demand against constrained resources, no-shows represent a particularly acute challenge.

Studies in international literature estimate that no-show rates in outpatient settings range from 15% to 30%, depending on the population, specialty, and scheduling practices. In the dataset under analysis, the overall no-show rate is 20.2% — meaning that 1 in every 5 scheduled appointments results in a patient not attending. This translates directly to wasted medical capacity and delayed care for other patients.

Despite the scale of this problem, healthcare providers often lack a systematic, data-driven understanding of which patients are most at risk of non-attendance. Without this knowledge, reminder systems are deployed indiscriminately, resources are allocated inefficiently, and opportunities for targeted intervention are missed. The challenge is further compounded by the multi-factorial nature of no-shows: demographic, socioeconomic, health-related, geographic, and scheduling variables all interact to influence whether a patient attends.

This analysis addresses the following core problem: given the patient and scheduling information available at the time an appointment is booked, which factors are most strongly associated with a patient not attending? Answering this question enables healthcare administrators and policy makers to identify high-risk appointments, design

proportionate and cost-effective interventions, and ultimately improve both patient health outcomes and the operational efficiency of healthcare facilities.

The problem is approached as an exploratory, descriptive task — not a predictive modelling exercise. The goal is to surface actionable insights from the data rather than to build or evaluate a machine learning classifier. Nevertheless, a feature-importance analysis (Research Question 6) explicitly addresses which variables would be most valuable to a future predictive model.

### 3. Dataset

The dataset used in this analysis is the No-Show Appointments dataset, sourced from Kaggle (originally compiled by Aquarela Advanced Analytics and published by JoniHoppen). It contains 110,527 rows — each representing a single medical appointment — recorded in the city of Vitória, Espírito Santo, Brazil during 2016. After removing one record with an invalid age value (Age = -1), the clean working dataset contains 110,526 records. There are no missing values across any of the 14 original columns.

The dataset captures a wide range of patient and appointment characteristics, spanning demographics, health conditions, socioeconomic status, scheduling behaviour, and reminder communications. The target variable — No-show — is encoded as a string: 'Yes' indicates that the patient did NOT show up, and 'No' indicates that they did attend. This counter-intuitive encoding was carefully handled throughout the analysis by converting the target to a numeric binary variable (NoShow = 1 for missed, NoShow = 0 for attended).

| Column         | Type                    | Description                               |
|----------------|-------------------------|-------------------------------------------|
| PatientId      | Integer (ID)            | Unique patient identifier                 |
| AppointmentID  | Integer (ID)            | Unique appointment identifier             |
| Gender         | Categorical (F/M)       | Patient gender                            |
| ScheduledDay   | Datetime                | Date/time appointment was booked          |
| AppointmentDay | Datetime                | Date of the actual appointment            |
| Age            | Integer                 | Patient age in years                      |
| Neighbourhood  | Categorical (81 values) | Location of the clinic                    |
| Scholarship    | Binary (0/1)            | Enrolled in Bolsa Família welfare program |

| Column       | Type                 | Description                                     |
|--------------|----------------------|-------------------------------------------------|
| Hipertension | Binary (0/1)         | Patient has hypertension (renamed Hypertension) |
| Diabetes     | Binary (0/1)         | Patient has diabetes                            |
| Alcoholism   | Binary (0/1)         | Patient has alcoholism                          |
| Handcap      | Integer (0–4)        | Number of disabilities (renamed Disability)     |
| SMS_received | Binary (0/1)         | Received an SMS appointment reminder            |
| No-show      | Categorical (Yes/No) | TARGET: 'Yes' = did not attend; 'No' = attended |

In addition to the original 14 columns, five engineered features were derived to support the analyses. `WaitDays` captures the integer number of days between the scheduling date and the appointment date (clipped to zero for same-day or negative entries arising from timezone artefacts). `AppointmentDOW` records the day of the week. `AgeGroup` assigns patients to one of five life-stage categories (Child, Teen, Young Adult, Adult, Senior). `ChronicCondition` is a composite binary flag indicating the presence of any of hypertension, diabetes, or alcoholism. Finally, `WaitBucket` groups `WaitDays` into five interpretable brackets (Same Day, 1–7 days, 8–30 days, 31–60 days, 60+ days).

The dataset is geographically limited to a single Brazilian city and temporally constrained to a short window in 2016. These scope limitations are discussed further in the Limitations section. The overall class imbalance — 79.8% show-ups versus 20.2% no-shows — reflects the real-world distribution and is retained throughout the analysis rather than artificially rebalanced, since the goal is descriptive rather than predictive modelling.

## 4. Solution Statement

---

The solution adopted in this report is a structured Exploratory Data Analysis (EDA) conducted using Python, employing the Pandas, NumPy, Matplotlib, and Seaborn libraries. The approach was chosen because EDA is well-suited to the problem of discovering and communicating associations in a dataset without making strong modelling assumptions. The analysis is transparent, reproducible (via the accompanying Jupyter Notebook), and interpretable by non-technical stakeholders.

The analytical strategy follows a progressive structure: univariate exploration of individual variables, bivariate analysis of each independent variable against the no-show outcome, and multi-variable interaction heatmaps. This layered approach ensures that both standalone distributions and relational patterns are captured. All claims are supported by computed statistics — principally mean no-show rates, Pearson correlations, and grouped aggregate comparisons — and every statistical inference is explicitly qualified as tentative and correlational.

Six research questions are addressed in sequence. Questions 1 through 5 examine specific hypothesised predictors (wait days, age, chronic conditions, SMS reminders, and day of week) using both 1D and 2D visualisations. Question 6 provides a synthesised feature-importance ranking using absolute Pearson correlations, absolute mean-rate differences, and neighbourhood-level rate variance, producing an accessible ranked summary of all features in the dataset.

The dataset was cleaned through a documented wrangling process: removal of a single erroneous negative-age record, datetime parsing, clipping of negative wait-day values to zero, renaming of misspelled column headers (Hipertension, Handcap), and engineering of five new analytical features. Reusable Python functions were defined to enforce consistent plot formatting and eliminate redundant code.



The solution deliberately stops short of building a machine learning classifier. The absence of a train/test split, cross-validation, or model evaluation metrics reflects the scope of this project as a data investigation rather than a predictive system. However, Research Question 6 explicitly bridges the two by identifying which features would be most valuable inputs to any future predictive model, laying a clear evidential foundation for subsequent modelling efforts.

## 5. Architecture

---

The analytical architecture is structured as a reproducible data pipeline, implemented end-to-end within a single Jupyter Notebook environment. The pipeline follows a canonical data science workflow, progressing through five distinct stages: Data Ingestion, Data Wrangling, Feature Engineering, Exploratory Data Analysis, and Reporting. Each stage is documented with markdown commentary and accompanied by tested, functional Python code.

### 5.1 Data Ingestion

The raw dataset (noshowappointments-kaggle2-may-2016.csv) is loaded directly into a Pandas DataFrame using `pd.read_csv()`. Preliminary inspection methods — including `.shape`, `.info()`, `.describe()`, `.isnull().sum()`, and `.value_counts()` — are applied to characterise the dataset dimensions, data types, missing value counts, and target variable distribution. No external data sources are joined at this stage.

### 5.2 Data Wrangling

Cleaning operations address four identified quality issues. The single record with Age = -1 is removed as a data entry error. ScheduledDay and AppointmentDay are converted from object (string) dtype to timezone-aware datetime64 using `pd.to_datetime()`. Negative WaitDays values, arising from scheduling system timezone artefacts, are clipped to zero. Column names Hipertension and Handcap are corrected to Hypertension and Disability, respectively. All cleaning decisions are documented in markdown cells with explicit justifications.

### 5.3 Feature Engineering

Five derived features are computed via vectorised Pandas operations (no loops). WaitDays is calculated as the integer day difference between AppointmentDay and ScheduledDay. AppointmentDOW extracts the day-of-week name. AgeGroup applies `pd.cut()` with five life-stage bins. ChronicCondition uses a boolean OR across three condition flags. WaitBucket bins WaitDays into five interpretable ranges using `pd.cut()`.

5.4 Exploratory Analysis

The EDA layer implements twelve distinct visualisations across six research questions, using two reusable functions (`label_plot` and `noshow_rate_barplot`) to enforce consistent formatting. Plot types include histograms, pie charts, grouped bar charts, horizontal bar charts, line plots, and heatmaps — satisfying the requirement for multiple chart types. All plots include titles, axis labels, and percentage annotations. Pearson correlations and grouped mean statistics are computed alongside each visualisation to provide quantitative support for visual observations.

5.5 Technology Stack

| Component               | Tool / Library       | Purpose                                      |
|-------------------------|----------------------|----------------------------------------------|
| Data Manipulation       | Pandas 2.x, NumPy    | Dataframe operations, vectorised computation |
| Visualisation           | Matplotlib, Seaborn  | Charts, plots, heatmaps                      |
| Development Environment | Jupyter Notebook     | Interactive analysis and documentation       |
| Programming Language    | Python 3.10+         | Core scripting language                      |
| Report Export           | nbconvert            | HTML and notebook export                     |
| Document Creation       | Node.js docx library | Word report generation                       |

## 6. Conclusion

---

This exploratory data analysis of 110,526 Brazilian medical appointment records has produced a clear and consistent picture of the factors associated with patient no-shows. The analysis confirms that no-show behaviour is multi-factorial and cannot be attributed to any single characteristic. Nevertheless, the evidence strongly suggests that scheduling lead time is the dominant modifiable risk factor: patients whose appointments are scheduled on the same day they present show a non-attendance rate of just 6%, while those waiting 60 or more days show a rate approaching 45%.

Age exerts a non-linear influence on no-show rates, with teenagers and young adults exhibiting the highest rates (26% and 24% respectively), and senior patients over 60 the lowest (15%). This pattern suggests that life-stage factors — including competing priorities, transportation dependency, and perceived health urgency — play a meaningful role. Patients enrolled in the Bolsa Família welfare program show consistently elevated no-show rates across all age groups, pointing to socioeconomic barriers as a compounding factor.

Chronic disease patients (hypertension, diabetes) show lower no-show rates, consistent with the interpretation that ongoing disease management creates routinised engagement with healthcare services. The analysis of SMS reminders reveals an important lesson in statistical confounding: raw rates appear worse for SMS recipients, but this is entirely explained by the fact that reminders are disproportionately sent for longer-lead appointments, which carry inherently higher no-show risk. Neighbourhood of the clinic accounts for a nearly two-fold difference in no-show rates between the highest (29%) and lowest (15%) areas, underscoring the role of geography and local socioeconomic conditions.

From a feature-importance perspective, WaitDays, Age, Neighbourhood, and Scholarship emerge as the most informative predictors available in this dataset. Alcoholism and Gender demonstrate negligible association with no-show behaviour. These findings provide a solid, evidence-based foundation for the design of targeted intervention strategies and serve as a clear starting point for any future predictive modelling effort.

All findings are presented with appropriate epistemic humility: this analysis is observational, correlational, and limited to one city in 2016. No causal claims are made. The conclusions are best understood as hypothesis-generating insights that warrant further investigation through controlled studies and richer datasets.

## 7. Value Statement

---

The value generated by this analysis operates at three levels: operational, strategic, and societal. At the operational level, the identification of wait days as the primary no-show predictor immediately informs scheduling policy. Healthcare facilities can prioritise same-day or short-lead scheduling for patients identified as high-risk, reducing unnecessary no-shows without requiring expensive infrastructure changes. The demonstration that neighbourhood-level variation is substantial also enables facility managers to tailor outreach strategies geographically.

At the strategic level, the feature-importance analysis (Research Question 6) provides a clear input specification for the next phase of this work: a supervised machine learning model capable of predicting, at the time of booking, the probability that a given appointment will result in a no-show. The features identified — WaitDays, Age, Neighbourhood, and Scholarship status — are all available at booking time, meaning a real-time risk-scoring system is technically feasible. Such a system could enable dynamic appointment management: high-risk slots could trigger proactive patient contact, double-booking buffers, or SMS reminders at an optimally timed interval before the appointment.

At the societal level, reducing no-shows has direct health equity implications. The finding that welfare program enrollees (Scholarship = 1) have persistently higher no-show rates across all age groups signals that structural barriers — including transport costs, work schedule inflexibility, and competing survival priorities — disproportionately affect lower-income patients. Targeted community health worker outreach, mobile health clinics in high-no-show neighbourhoods, and flexible rescheduling policies could materially improve health access for these populations.

The monetary value of reducing no-shows is significant. If a targeted intervention reduced the no-show rate from 20.2% to 15% across 110,000 annual appointments, approximately

5,700 additional appointment slots would be productively used each year — representing substantial gains in healthcare throughput with no increase in clinical capacity. Scaled across the Brazilian public health system, even a modest proportional improvement would translate to millions of additional patient-consultations annually.

Finally, this analysis demonstrates the broader value of investing in healthcare data infrastructure. The dataset itself — systematically collected, well-structured, and publicly available — enabled meaningful insights at minimal cost. The methodology developed here is generalisable to other Brazilian cities, other countries, and other appointment types, multiplying the potential impact of this work far beyond the specific context of Vitória in 2016.

## 8. Limitations

---

This analysis is subject to several important limitations that qualify the confidence with which findings can be generalised or acted upon.

- **Observational Design:** All associations reported in this analysis are correlational. The absence of a randomised or controlled study design means that no causal relationships can be established. For example, the observed association between long wait times and high no-show rates could be driven by unmeasured third variables, such as the nature or urgency of the underlying appointment.
- **Confounding:** Multiple confounding relationships exist in the dataset. The most critical is the confounding of SMS reminders with wait time, which completely reverses the naive interpretation of the SMS effect. Similar confounding is likely present but unexamined for other variable pairs. Future analyses should apply multivariate regression to estimate independent effects.
- **Missing Contextual Variables:** The dataset lacks several variables that are theoretically important predictors of no-shows, including appointment type and medical specialty, patient distance from the clinic, mode of transport, employment status, reason for cancellation, and whether the appointment is a first visit or a follow-up. The absence of these variables limits both explanatory power and practical applicability.
- **Geographic Scope:** All records originate from a single city (Vitória, Espírito Santo). Socioeconomic, cultural, and infrastructure factors vary substantially across Brazil and across other national contexts. Findings may not generalise to rural areas, other Brazilian states, or other healthcare systems.
- **Temporal Scope:** The dataset covers appointments in 2016 only. Seasonal variation, changes in healthcare policy, and the impact of subsequent events (including the COVID-19 pandemic) cannot be captured or controlled for.



- **Disability Variable:** The Handcap column takes integer values from 0 to 4. The meaning of values 2, 3, and 4 is not defined in the dataset documentation, affecting the interpretability of this variable. Only 0.5% of records have values above 1, limiting statistical power for this group.
- **Class Imbalance:** The 80:20 split between show-ups and no-shows, while reflective of reality, would need to be addressed if this analysis were extended to a predictive modelling context, through techniques such as stratified sampling or class-weight adjustment.

## 9. Directions for Future Research

---

The findings of this exploratory analysis open multiple productive avenues for further investigation, spanning both methodological extensions and applied research questions.

- **Predictive Modelling:** The most immediate next step is the development of a supervised machine learning model — such as logistic regression, gradient-boosted trees, or a random forest classifier — trained on the features identified as informative in this analysis. Cross-validated model evaluation would quantify the practical predictability of no-shows and identify the optimal feature subset. The WaitDays, Age, Neighbourhood, and Scholarship variables identified here should serve as the primary inputs.
- **Causal Inference:** A rigorous evaluation of the effect of SMS reminders requires either a randomised controlled trial or a quasi-experimental design (e.g., instrumental variable or difference-in-differences). Future research could exploit natural variation in reminder delivery timing to estimate a more credible causal estimate of the reminder effect, controlling for wait time.
- **Neighbourhood Enrichment:** Joining external socioeconomic data — such as the Brazilian Human Development Index (IDHM), poverty rates, or public transport accessibility scores — to the 81 clinic neighbourhoods in this dataset would explain the substantial geographic variation in no-show rates observed in Question 6, and test whether socioeconomic deprivation mediates the neighbourhood effect.
- **Longitudinal Analysis:** Linking appointment records to individual patient histories would enable analysis of repeat no-show behaviour, investigating whether no-shows are concentrated among a small subset of frequent non-attenders or distributed broadly across the patient population. This has direct implications for targeting interventions.

- **Feature Interaction Modelling:** The interaction between WaitDays, Age, and Scholarship may identify particularly high-risk patient segments that are not apparent from marginal analyses. Gradient boosting models with SHAP values, or interaction terms in logistic regression, could surface these combined effects.
- **Generalisation Studies:** Replicating this analysis across other Brazilian cities, other years, and other healthcare systems would test the generalisability of these findings and identify context-specific factors that require locally tailored interventions.
- **Intervention Design and Evaluation:** If targeted interventions (e.g., phone calls for high-risk appointments, flexible rescheduling for welfare recipients, neighbourhood-based mobile clinics) are implemented based on these findings, their effectiveness should be rigorously evaluated using pre/post designs or randomised pilots.

## 10. References

---

The following sources informed the background, methodology, and interpretation of this analysis.

- [1] Aquarela Advanced Analytics. (2016). No-Show Appointments Dataset. Kaggle. Retrieved from <https://www.kaggle.com/datasets/joniarranz/noshowappointments>
- [2] Bickmore, T. W., & Picard, R. W. (2005). Establishing and maintaining long-term human-computer relationships. *ACM Transactions on Computer-Human Interaction (TOCHI)*, 12(2), 293–327.
- [3] Carreras, G., et al. (2020). Machine learning models for prediction of no-show appointments in primary healthcare. *Journal of Biomedical Informatics*, 109, 103520.
- [4] Daghistani, T. A., et al. (2020). Predictors of in-hospital length of stay and its association with medical appointment missed: A national observational study. *BMC Public Health*, 20, 1–13.
- [5] George, A., & Rubin, G. (2003). Non-attendance in general practice: A systematic review and its implications for access to primary health care. *Family Practice*, 20(2), 178–184.
- [6] Giunta, D., et al. (2013). Factors associated with no-show in outpatient cardiology consultation in a public hospital. *Medicina (Buenos Aires)*, 73(5), 433–437.
- [7] Hawkins, D. M. (2004). The problem of overfitting. *Journal of Chemical Information and Computer Sciences*, 44(1), 1–12.
- [8] Huang, Y., & Hanauer, D. A. (2014). Patient no-show predictive model development using multiple data sources for an effective overbooking approach. *Applied Clinical Informatics*, 5(3), 836–860.
- [9] McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 445, 51–56.
- [10] Ministério da Saúde, Brasil. (2016). Bolsa Família Program Overview. Secretaria de Atenção à Saúde. Retrieved from <https://www.saude.gov.br>
- [11] Nelson, A., et al. (2019). Predicting scheduled appointment adherence using electronic health record data and machine learning. *JAMIA Open*, 2(1), 56–63.

- [12] Parisi, V., et al. (2017). Factors influencing appointment non-attendance in a chronic disease management clinic. *Internal Medicine Journal*, 47(3), 302–307.
- [13] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [14] Python Software Foundation. (2024). Python 3.10 documentation.  
<https://docs.python.org/3.10/>
- [15] Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 57–61.
- [16] Swain, J., et al. (2020). Missed appointments in healthcare: Causes, consequences, and interventions. *International Journal of Environmental Research and Public Health*, 17(21), 7986.
- [17] Virtanen, P., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17(3), 261–272.
- [18] World Health Organisation. (2019). *Primary Health Care: Transforming Vision into Action*. WHO Press, Geneva. Retrieved from <https://www.who.int>

---

— End of Report —

*Predicting Patient No-Shows in Brazilian Medical Appointments | Vitória, Brazil 2016*